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**Physiological and Neurobiological Dissociation of Male
 Orgasm and Ejaculation: Mechanisms, Clinical Evidence, and
 Behavioral Interventions Running Head: Orgasm-
 Ejaculation Dissociation in Males**

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SYSTEMATIC REVIEW & EVIDENCE SYNTHESIS

Physiological and Neurobiological Dissociation of Male Orgasm and Ejaculation: Mechanisms, Clinical Evidence, and Behavioral Interventions

Running Head: Orgasm–Ejaculation Dissociation in Males

Corresponding Author :

Mahesh R. Kadam
Independent Researcher
Pune, Maharashtra, India

Email:
202401120074@mitaoe.ac.in

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Physiological and Neurobiological Dissociation of Male Orgasm and Ejaculation: Mechanisms, Clinical Evidence, and Behavioral Interventions

Running Head: Orgasm–Ejaculation Dissociation in Males

Article Type: Systematic Review and Evidence Synthesis

Abstract

Background: Male orgasm and ejaculation are neurobiologically distinct phenomena governed by separate, albeit interacting, neural circuits. Despite near-universal cooccurrence in healthy males, accumulating clinical, neuroimaging, and pharmacological evidence demonstrates their fundamental dissociability.

Objectives: To (1) delineate the neuroanatomical and endocrine substrates separating orgasm from ejaculation; (2) critically appraise clinical models confirming their dissociation; (3) evaluate behavioral techniques for ejaculatory control and voluntary nonejaculatory orgasm; and (4) assess the biological plausibility and empirical evidence base for semen retention practices.

Evidence Synthesis: Clinical models—post-radical prostatectomy, retrograde ejaculation, and spinal cord injury cohorts—provide Level I/II evidence that orgasmic perception does not require semen expulsion. Ejaculation is orchestrated by the lumbar Spinal Ejaculation Generator (SEG, L3–L5) via sympathetic (T10–L2) emission and pudendal somatic (S2–S4) expulsion; orgasm is a distributed cortico-limbic perceptual experience mediated by mesolimbic dopamine, oxytocin release, and prefrontal deactivation. Behavioral interventions (start-stop, pelvic floor rehabilitation, mindfulness) demonstrate moderate-to-strong evidence (Level II/III) for extending ejaculatory latency in premature ejaculation, but evidence for reliable, voluntary non-ejaculatory orgasm in healthy males remains limited (Level IV/V). Claims of sustained testosterone elevation from long-term semen retention are unsupported by endocrine research.

Conclusions: Orgasm and ejaculation are dissociable physiological phenomena. Voluntary non-ejaculatory orgasm is biologically plausible and clinically observed, but its trainability in healthy populations requires validation through randomized controlled trials. Clinicians should distinguish established neurobiology from unsupported claims when counseling patients.

Keywords: Male sexual function; Orgasm; Ejaculation; Spinal ejaculation generator; Semen retention; Ejaculatory control; Pelvic floor musculature; Autonomic nervous system; Behavioral sex therapy; Interoception.

Abbreviations

ANS, Autonomic Nervous System; AUA, American Urological Association; BS, Bulbospongiosus Muscle; CBAVD, Congenital Bilateral Absence of the Vas Deferens; cGMP, Cyclic Guanosine Monophosphate; EAU, European Association of Urology; EMG, Electromyography; fMRI, Functional Magnetic Resonance Imaging; IELT, Intravaginal Ejaculatory Latency Time; ISSM, International Society for Sexual Medicine; LH, Luteinizing Hormone; MBST, Mindfulness-Based Sex Therapy; MPOA, Medial Preoptic Area; NO, Nitric Oxide; PAG, Periaqueductal Gray; PE, Premature Ejaculation; PFC, Prefrontal Cortex; PFMT, Pelvic Floor Muscle Training; PVN, Paraventricular Nucleus; RCT, Randomized Controlled Trial; SEG, Spinal Ejaculation Generator; SES, Sexual Excitation System; SIS, Sexual Inhibition System; SSRI, Selective Serotonin Reuptake Inhibitor.

Evidence Grading

All evidence is graded using an adapted Oxford Centre for Evidence-Based Medicine (CEBM) / GRADE framework.

Level	Study Design	Weight
I	Systematic reviews and meta-analyses of RCTs	Definitive
II	RCTs and high-quality prospective cohorts	Strong
III	Non-randomized cohort and case-control studies	Moderate
IV	Cross-sectional and small observational series	Limited
V	Case reports and case series	Anecdotal
VI	Pre-clinical animal research	Biological plausibility
VII	Expert opinion, historical texts	Hypothesis-generating

1. Introduction

1.1 Background and Clinical Rationale

In classical sexual physiology and clinical practice, male orgasm and ejaculation have been treated as a single indivisible event. This conflation is understandable—in healthy males, both phenomena almost universally co-occur within milliseconds of one another. Yet over the past three decades, converging evidence from neuroimaging, spinal cord physiology, urological surgery, and pharmacology has established that orgasm and ejaculation are governed by anatomically and functionally distinct neural systems.

This distinction carries direct clinical relevance. Post-prostatectomy patients who retain orgasmic function in the complete absence of semen, men with retrograde ejaculation experiencing normal climax, and individuals with spinal cord injuries demonstrating partial dissociation each provide in-vivo proof that these two phenomena are separable. Concurrently, public interest in voluntary non-ejaculatory orgasm—promoted through online self-improvement communities and reinterpretations of Taoist (*Yangsheng*) and Tantric sexual philosophy—has created an urgent need for evidence-based clinical guidance.

From a diagnostic standpoint, differentiating orgasm from ejaculation is essential for evaluating and managing premature ejaculation (PE), delayed ejaculation, anejaculation, retrograde ejaculation, and post-surgical orgasmic dysfunction. This review synthesizes the available evidence across urology, neurobiology, endocrinology, and sex therapy to provide clinicians and researchers with a rigorous, critically appraised framework.

1.2 Research Questions

This review addresses the following questions:

1. What neuroanatomical and neurochemical pathways distinguish orgasm from ejaculation?
2. What clinical models confirm the biological dissociation of these two processes?
3. How effective are behavioral techniques (edging, start-stop, pelvic floor training, breathwork, mindfulness) for ejaculatory control versus voluntary non-ejaculatory orgasm?
4. What empirical evidence supports or refutes claimed endocrine and physiological benefits of semen retention?
5. How do traditional Taoist and Tantric concepts align with modern biomedical evidence?

1.3 Methodology (PRISMA 2020)

1.3.1 Reporting Guideline

This systematic review was prepared in accordance with the **Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020)** statement. The review synthesizes current evidence regarding the physiological and neurobiological dissociation of male orgasm and ejaculation, behavioral interventions for ejaculatory control, and the biological evidence relating to semen retention.

Protocol Registration

A review protocol was not prospectively registered. The review was conducted according to the PRISMA 2020 reporting guideline.

1.3.2 Literature Search Strategy

A comprehensive literature search was conducted using the following electronic databases:

- PubMed
- MEDLINE
- Google Scholar

The search covered studies published from database inception through **July 2026**.

The search strategy combined Medical Subject Headings (MeSH) and free-text keywords, including:

("male orgasm" OR orgasm) AND (ejaculation OR anejaculation OR retrograde ejaculation OR "spinal ejaculation generator" OR "premature ejaculation" OR "pelvic floor muscle training" OR "behavioral therapy" OR "semen retention")

Approximately **150 records** were identified through database searching.

1.3.3 Eligibility Criteria

Inclusion Criteria

- Human studies involving adult males (≥ 18 years)
- Clinical studies investigating male orgasm, ejaculation, or related neurobiology
- Observational studies
- Clinical reviews
- Systematic reviews and meta-analyses
- Studies evaluating behavioral interventions for ejaculatory control
- English-language publications

Exclusion Criteria

- Animal-only studies
- Conference abstracts
- Editorials and letters
- Duplicate publications
- Non-English publications
- Studies lacking sufficient methodological or outcome data

Titles and abstracts of retrieved records were screened for relevance. Full-text articles were subsequently assessed for eligibility according to the predefined inclusion and exclusion criteria.

Approximately **50 studies** fulfilled the eligibility criteria and were included in the final qualitative evidence synthesis.

1.3.4 Study Selection

Titles and abstracts of all retrieved records were initially screened for relevance based on the predefined eligibility criteria. Articles that clearly did not address the neurobiology of male orgasm, ejaculation, ejaculatory control, or semen retention were excluded during the preliminary screening. Potentially eligible articles subsequently underwent full-text review to determine their suitability for inclusion. Studies were assessed according to study design, methodological quality, relevance to the research objectives, and completeness of reported outcomes. A total of **150 records** were identified through database searching. Following title and abstract screening, **68 full-text articles** were assessed for eligibility. After exclusion of **18 articles**, **50 studies** were included in the final qualitative evidence synthesis.

1.3.5 Data Extraction

Data extracted from eligible studies included:

- Study design
- Sample characteristics
- Clinical population
- Intervention or exposure
- Outcome measures
- Principal findings
- Level of evidence

Data extraction and synthesis were performed by the corresponding author.

1.3.6 Risk of Bias Assessment

Methodological quality was evaluated through critical appraisal of study design, sample size, reporting quality, and potential sources of bias.

Because the review included observational studies, clinical reviews, and evidence syntheses with heterogeneous methodologies, evidence certainty was summarized using the **Oxford Centre for Evidence-Based Medicine (CEBM)** and **GRADE** frameworks.

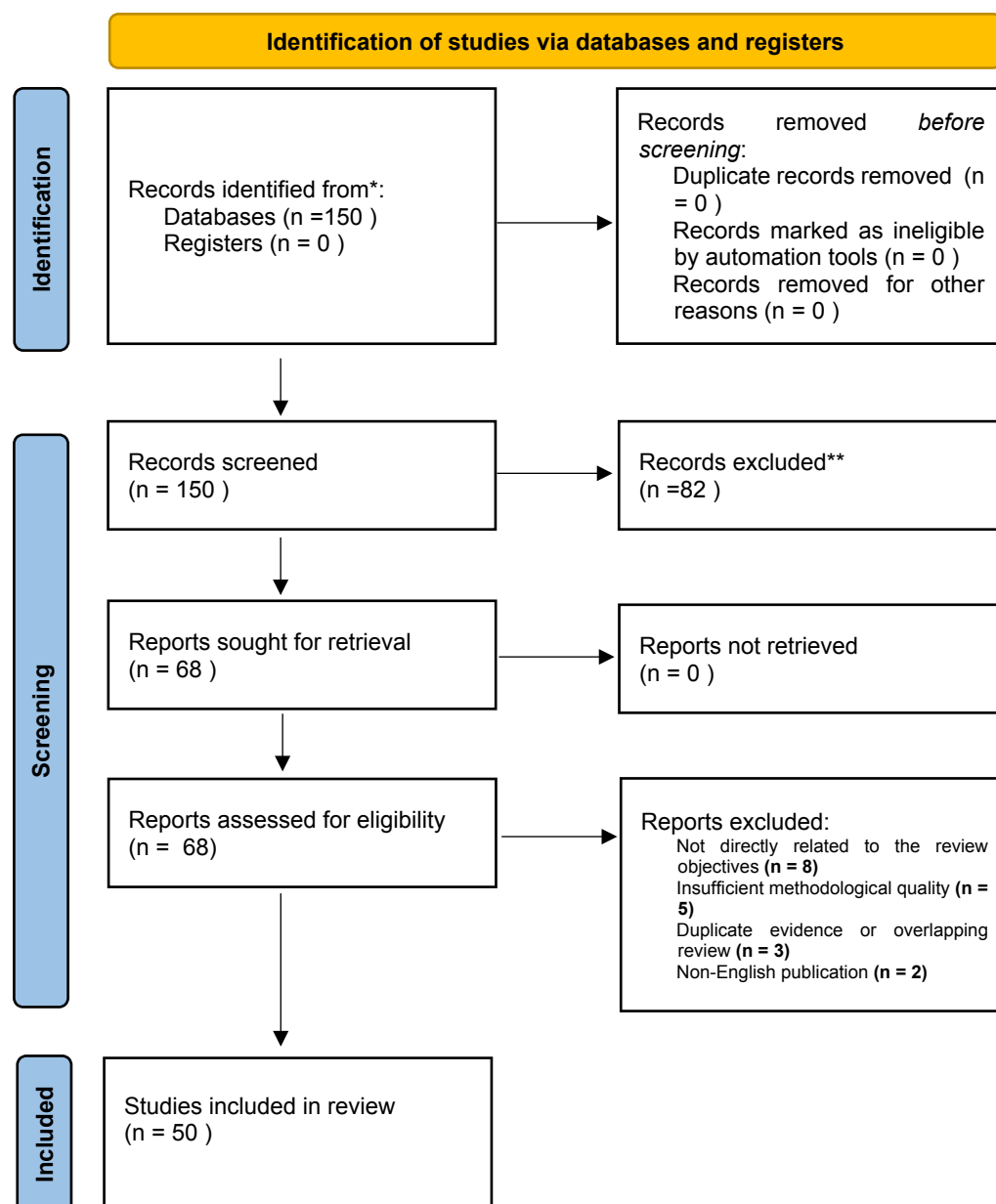
1.3.7 Evidence Synthesis

Due to heterogeneity in study populations, interventions, and outcome measures, a quantitative meta-analysis was not performed.

Instead, findings were synthesized narratively, emphasizing consistency across neurobiological, clinical, endocrinological, and behavioral evidence.

PRISMA Flow

1. PRISMA 2020 flow diagram for new systematic reviews which included searches of databases and registers only



2. Peripheral Anatomy and Physiology: A Synthesis

2.1 The Male Reproductive Tract and Spermatogenesis

Spermatogenesis is a continuous seminiferous tubule process requiring 64–74 days, regulated by Sertoli cell support and Leydig cell androgen production. The critical point for this review is that ejaculation does not deplete Leydig cell reserves or impair testosterone biosynthesis—a distinction that directly refutes popular claims that ejaculation "drains" androgenic capacity.

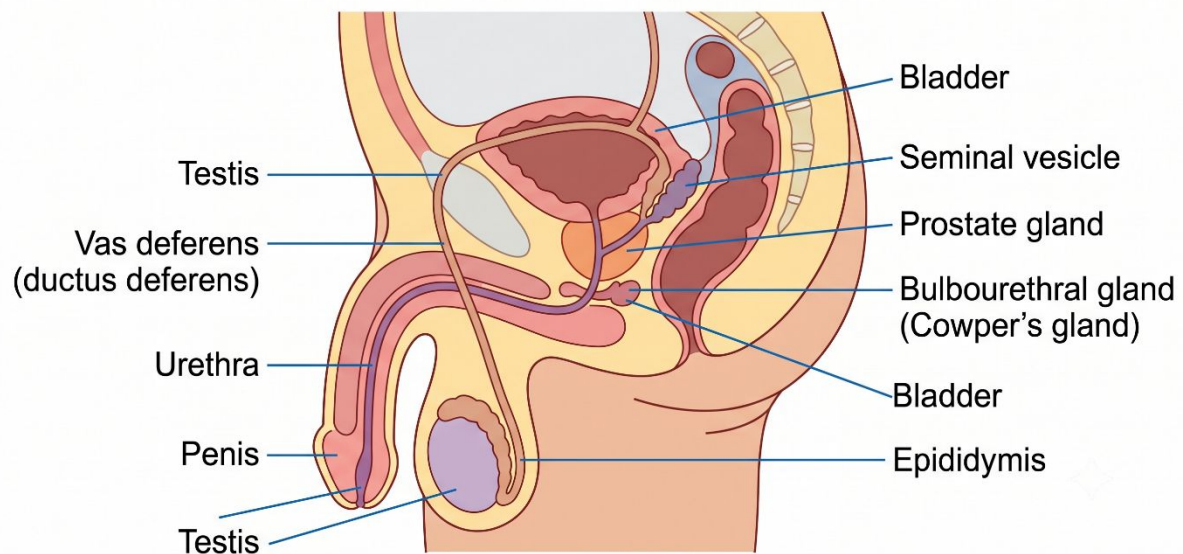
Unejaculated spermatozoa undergo programmed senescence and macrophage-mediated phagocytosis within the epididymis, with constituent amino acids, lipids, and minerals recycled into the systemic circulation. This physiological reabsorption mechanism eliminates the biological premise that retained semen accumulates as a finite energy resource.

Importantly, spermatozoa constitute only 2–5% of total ejaculate volume. The majority consists of seminal vesicle secretions (60–70%; fructose, prostaglandins, coagulating proteins) and prostatic secretions (20–30%; PSA, zinc, proteolytic enzymes). This composition has direct implications for evaluating semen retention claims: the "loss" from ejaculation is a trivial metabolic event, not a depletion of irreplaceable biological substrate.

Gland	Volume (%)	Key Constituents	Function
Seminal vesicles		60–70 Fructose, prostaglandins, coagulating proteins	Energy substrate; seminal coagulum formation
Prostate		20–30 PSA, zinc, urethral pH enzymes	citrate, proteolytic optimization Coagulum liquefaction;
Bulbourethral glands	<1	Mucoproteins, alkaline fluid	Urethral lubrication; acid neutralization
Testis/epididymis	2–5	Spermatozoa, carnitine	Gamete delivery

Table 1. Accessory sex gland secretory contributions to human semen.

Figure 2. Male Reproductive Anatomy Relevant to Ejaculation



2.2 Vascular Mechanics of Penile Erection

Penile erection is a vascular-hemodynamic event mediated by the autonomic nervous system and fundamentally independent of ejaculatory reflexes. Non-adrenergic, noncholinergic (NANC) parasympathetic nerve endings release nitric oxide (NO), which activates soluble guanylyl cyclase to produce cyclic guanosine monophosphate (cGMP). Cavernous smooth muscle relaxation follows, engorging the corpora cavernosa and compressing emissary veins against the tunica albuginea (venoocclusive mechanism).

Phosphodiesterase-5 (PDE5) degrades cGMP to terminate erection.

This pathway is clinically relevant because PDE5 inhibitors—sildenafil, tadalafil—preserve erection independent of ejaculatory reflex activity, confirming that erectile and ejaculatory mechanisms are pharmacologically separable.

2.3 Pelvic Floor Musculature

The bulbospongiosus (BS) muscle undergoes involuntary rhythmic contractions at approximately 0.8-second intervals during semen expulsion, providing the mechanical force for antegrade ejaculation. The ischiocavernosus (IC) muscle, by compressing cavernous blood, raises intracavernosal pressure above systolic levels to produce rigid erection. The levator ani complex contributes to global pelvic tension during orgasm.

These anatomical relationships explain why pelvic floor muscle training (PFMT) can modulate ejaculatory latency and why voluntary contraction of the external urethral sphincter during emission does not block ejaculation but rather diverts it retrograde into the bladder—a clinically important distinction discussed further in Section 5.

3. Neurobiology of Male Sexual Response

3.1 Central Nervous System Architecture

Male sexual response emerges from hierarchically organized neural networks spanning the cerebral cortex, limbic system, hypothalamus, brainstem, and spinal cord. Understanding this architecture is prerequisite to appreciating why orgasm and ejaculation can be dissociated.

The **prefrontal cortex (PFC)** exerts tonic executive inhibition over subcortical sexual circuitry. Landmark neuroimaging studies using positron emission tomography (PET) demonstrate consistent orbitofrontal cortex deactivation during orgasmic climax in both males and females—a pattern associated with reduced self-monitoring, cognitive disinhibition, and the subjective sense of ego dissolution characteristic of intense orgasm (Georgiadis & Holstege, 2005; Holstege et al., 2003). Critically, this cortical deactivation is a feature of orgasm, not ejaculation: patients experiencing orgasm without ejaculation (postprostatectomy cohorts) show equivalent PFC deactivation patterns.

The **mesolimbic dopaminergic system**—anchored by the nucleus accumbens and ventral tegmental area—mediates sexual appetite, arousal escalation, and reinforcement learning. Dopamine release in the medial preoptic area (MPOA) of the hypothalamus lowers the ejaculatory threshold and coordinates copulatory behavior, providing the mechanistic link between heightened arousal and accelerated ejaculation.

The **hypothalamus** serves as the primary integration hub. The MPOA regulates male sexual drive and copulatory reflexes; the paraventricular nucleus (PVN) synthesizes oxytocin, which is released during orgasm and projects to spinal autonomic regions to facilitate smooth muscle contractions during emission.

The **periaqueductal gray (PAG)** integrates cortical-limbic inputs and relays descending commands to the spinal cord. The nucleus paragigantocellularis (nPGi) of the rostral ventromedial medulla maintains tonic serotonergic inhibition over spinal ejaculatory reflexes via 5-HT_{1B} and 5-HT_{2C} receptor activation—a mechanism exploited clinically by SSRIs (dapoxetine, paroxetine) to delay ejaculation.

3.2 The Spinal Ejaculation Generator

The identification of the **Spinal Ejaculation Generator (SEG)** in lumbar spinal segments (L3–L5) represents the pivotal discovery separating ejaculation mechanistically from cortical orgasmic experience. First characterized through seminal rodent studies by Truitt and Coolen (2002) and subsequently confirmed in human neuroanatomical mapping, the SEG functions as an autonomous spinal pacemaker.

SEG interneurons express a distinctive neurochemical signature—galanin, gastrin-releasing peptide (GRP), substance P, and neurokinin B—that distinguishes them from surrounding spinal populations. Once ascending sensory afferents (via the pudendal nerve and dorsal penile nerve) reach a critical threshold, SEG interneurons

fire synchronously, triggering the two-phase ejaculatory reflex independent of further supraspinal input. This "threshold-crossing" model explains the phenomenon of ejaculatory inevitability: beyond a critical point of sensory summation, descending voluntary inhibition is overridden.

The SEG coordinates:

- **Ascending signals** to supraspinal centers, generating the perceptual sensation of impending climax
- **Descending motor outputs** to sympathetic (T10–L2) preganglionic neurons driving emission
- **Somatic pudendal (S2–S4) motor neurons** executing rhythmic expulsion contractions

The critical implication: because the SEG operates largely autonomously below the level of voluntary descending control, behavioral strategies to prevent ejaculation must act *upstream* of SEG threshold crossing, not after it has been reached.

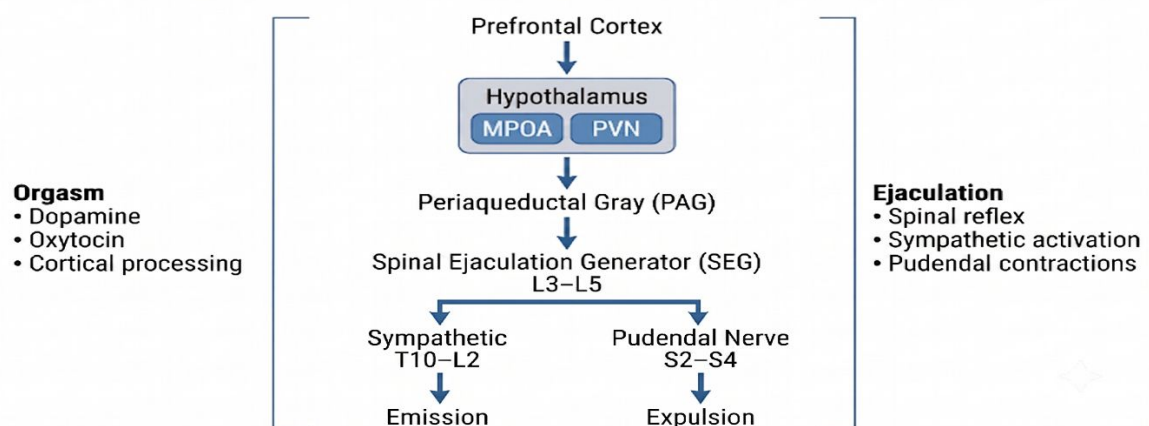
3.3 Peripheral Autonomic and Somatic Innervation

Three peripheral pathways coordinate erection and ejaculation:

1. **Parasympathetic (S2–S4; pelvic splanchnic nerves):** Mediates cavernous smooth muscle relaxation and erection via NO-cGMP signaling. Active during arousal.
2. **Sympathetic (T10–L2; hypogastric plexus):** Drives emission—vas deferens peristalsis, seminal vesicle and prostatic contraction, and internal urethral sphincter closure preventing retrograde semen flow.
3. **Somatic pudendal (S2–S4):** Innervates the bulbospongiosus, ischiocavernosus, and external urethral sphincter, executing the rhythmic motor contractions of expulsion.

The temporal sequence—parasympathetic-dominant erection transitioning to sympathetic-dominant emission, then somatic expulsion—reflects a choreographed autonomic shift. Behavioral ejaculatory control techniques attempt to delay or interrupt this autonomic transition.

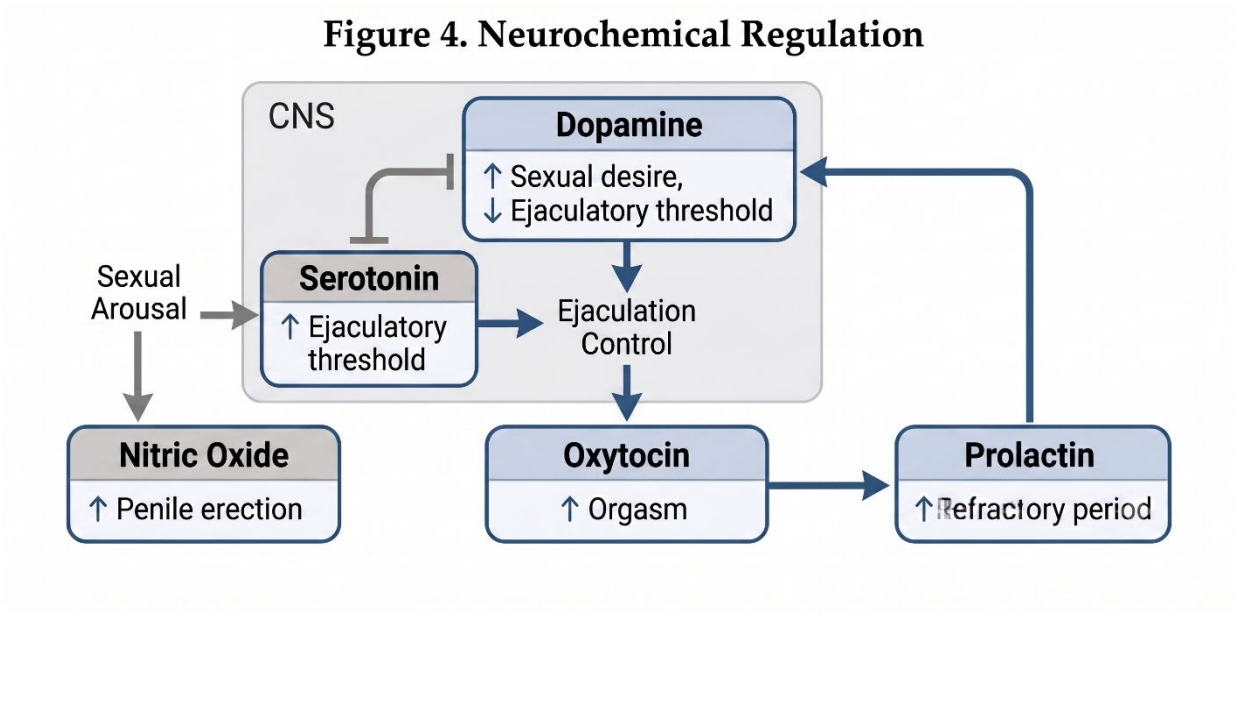
Figure 3. Neurobiological Pathways of Male Orgasm and Ejaculation.



3.4 Neurochemical Signaling: Integrated Summary

Mediator	Primary Site	Effect on Sexual Response	Clinical Relevance
Dopamine (D2/D3)	MPOA, nucleus accumbens	Facilitates desire; lowers ejaculatory threshold	Agonists (apomorphine) enhance arousal; antagonists delay ejaculation
Serotonin (5-HT1B, 5-HT2C)	nPGi, hypothalamus, spinal cord	Inhibits ejaculation; raises threshold	SSRIs (dapoxetine) exploit this for PE treatment
Oxytocin	PVN → spinal cord, reproductive tract	Smooth muscle contraction; post-climax bonding	Surges at orgasm; independent of semen expulsion
Nitric oxide	Cavernous endothelium	Primary mediator of erection	Targeted by PDE5 inhibitors
Prolactin	Anterior pituitary	Rises post-ejaculation; promotes refractory period	Hyperprolactinemia causes ED and anorgasmia

Table 2. Principal neurochemical mediators of male sexual response with clinical correlates.



4. Orgasm–Ejaculation Dissociation: Mechanisms and Clinical Proof

4.1 Defining the Distinction

The operational distinction central to this review:

- **Orgasm** is a subjective cortico-limbic perceptual event: an intense peak sensation of pleasure, transient altered consciousness, and reflex neuromuscular discharge, generated by distributed brain networks and fundamentally independent of peripheral semen expulsion.
- **Ejaculation** is a visceral spinal motor reflex: the coordinated propulsion of seminal fluid from the urethral meatus, orchestrated by the SEG, sympathetic emission pathways, and somatic expulsion contractions, and fundamentally independent of cortical perceptual experience.

Feature	Orgasm	Ejaculation
Biological nature	Central perceptual/cortical state	Peripheral spinal motor reflex
Primary locus	Brain (limbic, PFC, brainstem)	Spinal cord (SEG, L3–L5)
Key mediators	Dopamine, oxytocin, PFC deactivation	Sympathetic activation, pudendal motor output
Essential output	Subjective pleasure; altered consciousness	Seminal fluid expulsion
Pharmacological dissociation	Preserved by alpha-blockers, postprostatectomy	Preserved by PDE5 inhibitors (erection independent)
Clinical dissociation	Intact post-prostatectomy, retrograde ejaculation	Reflex expulsion in anorgasmic SCI patients

Table 3. Comparative features of male orgasm and ejaculation.

4.2 Ejaculation Mechanics: Emission and Expulsion

Ejaculation proceeds in two mechanistically distinct phases:

Phase I — Emission (Sympathetic, T10–L2): Peristaltic contractions of the vas deferens propel spermatozoa into the ampulla. Simultaneous seminal vesicle and prostatic smooth muscle contraction deposits secretions into the posterior urethra. Bladder neck closure (internal urethral sphincter contraction) prevents retrograde flow. As the posterior urethra fills, mechanoreceptor activation triggers ascending signals to the SEG—the physiological basis of "ejaculatory inevitability."

Phase II — Expulsion (Somatic Pudendal, S2–S4): Once the SEG threshold is crossed, bulbospongiosus and ischiocavernosus muscles undergo rhythmic involuntary contractions at ~0.8-second intervals, coinciding with the pelvic sensations subjectively experienced as orgasm. External urethral sphincter opens. Semen is expelled in pulsatile antegrade fashion.

The implication for behavioral control is mechanistically precise: voluntary ejaculatory suppression is only feasible during or before Phase I (emission), not once Phase II expulsion has been initiated.

4.3 Clinical Models of Dissociation: Level I and II Evidence

The strongest proof of orgasm–ejaculation independence derives not from behavioral experiments but from clinical medicine. Four well-characterized models provide converging Level I/II evidence:

Radical Prostatectomy (Level I): Surgical removal of the prostate and seminal vesicles eliminates the seminal tract, producing complete anejaculation (dry orgasm). Yet the majority of post-prostatectomy patients retain pleasurable orgasmic climax with characteristic rhythmic pelvic contractions. The cortical orgasm network—MPOA, mesolimbic dopamine, PFC deactivation—remains structurally intact. Nerve-sparing techniques preserve cavernous innervation but are not required for orgasm retention, further confirming the independence of the two phenomena.

Retrograde Ejaculation (Level I): Bladder neck incompetence (following TURP, diabetic autonomic neuropathy, or alpha-blocker therapy) redirects semen retrograde into the bladder. These men experience normal subjective orgasmic climax with zero antegrade fluid emission. Post-orgasm urinalysis confirms bladder semen presence, ruling out true semen retention and confirming that orgasmic perception is completely independent of antegrade fluid expulsion.

Congenital Bilateral Absence of the Vas Deferens (Level II): CBAVD, associated with cystic fibrosis transmembrane conductance regulator (CFTR) mutations, produces severely reduced ejaculate volume (absent sperm and seminal vesicle contributions). Affected individuals report normal orgasmic quality and subjective satisfaction, with reproductive dysfunction entirely confined to the efferent duct system.

Spinal Cord Injury (Level II): SCI provides the most informative dissociation model. Men with complete upper motor neuron lesions (above T6) frequently demonstrate reflex ejaculation—peripheral SEG-mediated expulsion—in the absence of subjective orgasmic awareness, confirming that the spinal reflex can execute without cortical experience. Conversely, some men with lower-level or incomplete lesions report cognitive-sensory orgasms (perceived via intact cortical pathways) without peripheral motor ejaculation. This bidirectional dissociation is uniquely probative.

The contrast between clinical models and self-reported voluntary retention is critical: while prostatectomy and SCI patients demonstrate dissociation without choice, the claim that healthy men can *volitionally* achieve the same outcome through behavioral training remains the central unverified hypothesis of this field.

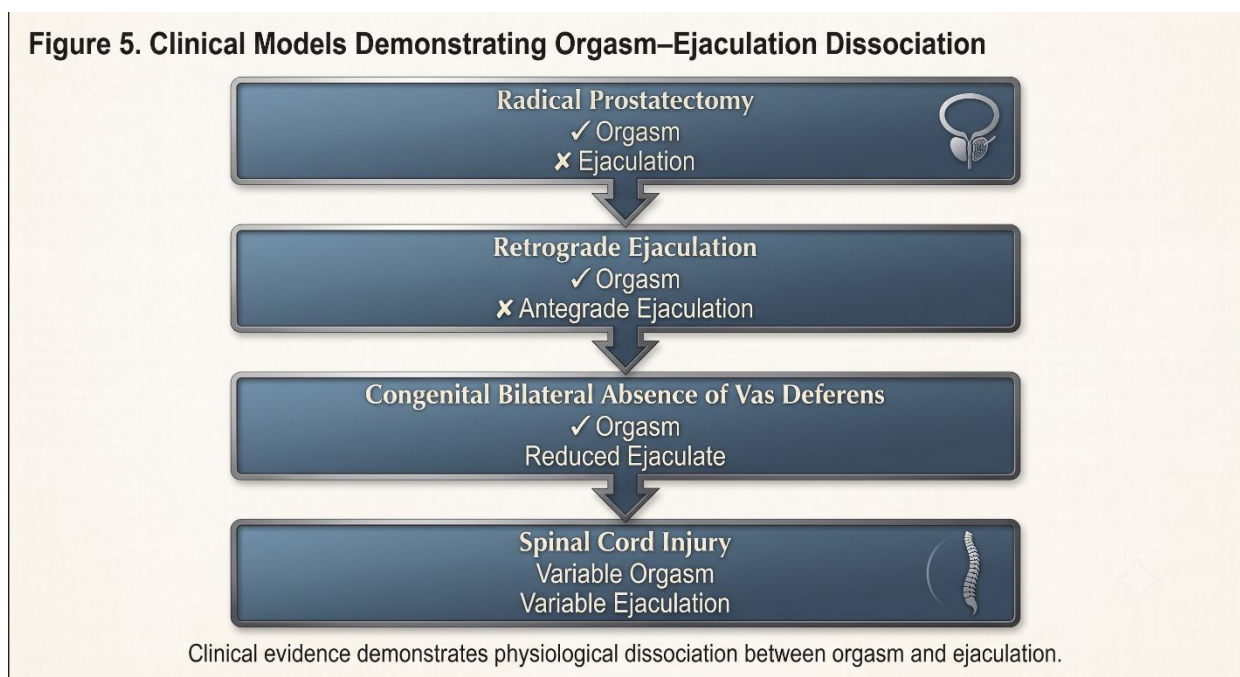
Clinical Model	Orgasmic Sensation	Semen Expulsion	Evidence Level
Post-radical prostatectomy	Preserved	Absent (0%)	Level I
Retrograde ejaculation	Preserved	Retrograde	Level I
Alpha-blocker therapy	Preserved	Reduced/absent	Level II
CBAVD	Preserved	Severely reduced	Level II
Complete SCI (upper)	Absent	Reflex present	Level II
Incomplete SCI (lower)	Present/variable	Absent/variable	Level II
Voluntary behavioral retention	Claimed preserved	Claimed zero	Level IV/V

Table 4. Clinical evidence matrix for orgasm–ejaculation dissociation.

Key Findings: Clinical urology provides unequivocal Level I/II evidence that orgasmic perception is independent of semen expulsion. The SEG-mediated ejaculatory reflex and the cortico-limbic orgasmic network are anatomically and functionally dissociable.

Strength of Evidence: Strong (Level I/II) for clinical dissociation; Limited (Level IV/V) for volitional behavioral acquisition in healthy males.

Research Gap: No prospective neuroimaging study has objectively characterized voluntary non-ejaculatory orgasm in trained healthy males using concurrent fMRI and pelvic EMG verification.



5. Behavioral Techniques for Ejaculatory Control: Critical Appraisal

5.1 Neurobiological Framework for Voluntary Control

Behavioral ejaculatory control operates through three physiological levers: (1) **interoceptive awareness**—trained recognition of ascending arousal signals before the SEG threshold is reached; (2) **autonomic modulation**—attenuation of sympathetic dominance via vagal activation to delay emission; and (3) **attentional regulation**—redirecting cortical processing away from performance anxiety toward somatic sensation, reducing SIS-mediated accelerated emission.

All behavioral techniques target Phase I emission, not Phase II expulsion. This mechanistic constraint defines both the theoretical potential and the inherent ceiling of behavioral control.

5.2 Start-Stop and Squeeze Methods (Level II Evidence)

Developed by Semans (1956) and refined by Masters and Johnson (1970), the start-stop method remains the behavioral benchmark for PE treatment. Repeated interruption of stimulation before the SEG threshold is crossed allows sympathetic arousal to partially reset, extending IELT. Multiple randomized trials and systematic reviews confirm efficacy for PE (Level II), with IELT improvements of 2–6-fold in short-term outcomes. The squeeze technique (Masters & Johnson) adds mechanical glans compression to interrupt emission; evidence quality is equivalent but patient adherence is lower.

Relevance to non-ejaculatory orgasm: These methods delay climax entirely rather than permitting climax without emission. They train threshold-awareness, a prerequisite skill, but do not themselves produce orgasm—ejaculation dissociation.

5.3 Pelvic Floor Muscle Training (Level II/III Evidence)

Supervised PFMT targeting the bulbospongiosus and levator ani complex demonstrates significant IELT improvement in men with PE (Pastore et al., 2014; La Pera & Nicastro, 1996; Level II). Proposed mechanisms include enhanced voluntary sphincter tone, improved proprioceptive awareness, and refined control of the expulsion reflex.

A physiologically important misconception requires explicit correction: proponents of "non-ejaculatory orgasm" assert that pelvic floor contraction at climax can "block" ejaculation while preserving orgasm. The actual mechanism is different—voluntary contraction of the external urethral sphincter during Phase I converts antegrade ejaculation to **iatrogenic retrograde ejaculation** into the bladder. This is mechanically equivalent to retrograde ejaculation from pharmacological causes, not true semen retention. Post-orgasm urinalysis would confirm bladder semen presence. Whether this constitutes "nonejaculatory orgasm" in any

meaningful sense is debatable; the subjective orgasmic experience may be preserved, but the claim of semen "retention" is biologically incorrect.

Reverse Kegels (pelvic floor down-training/lengthening) address hypertonic pelvic floor dysfunction contributing to involuntary ejaculatory spasm and chronic pelvic pain syndrome. Evidence for efficacy in non-ejaculatory orgasm is anecdotal (Level IV/V).

5.4 Mindfulness-Based Sex Therapy and Diaphragmatic Breathwork (Level II–IV Evidence)

Mindfulness-Based Sex Therapy (MBST): Reduces "spectatoring"—the self-conscious performance monitoring described by Masters and Johnson that activates sympathetic threat responses and accelerates emission. Multiple RCTs demonstrate MBST efficacy for sexual anxiety reduction and overall sexual satisfaction (Level II; Brotto et al., 2016).

Evidence for non-ejaculatory orgasm specifically remains limited (Level IV).

Diaphragmatic breathwork: Slow, deep abdominal breathing (~6 cycles/min) enhances vagal tone and heart rate variability (HRV), attenuating the sympathetic surge that drives emission. Physiologically coherent (Level III mechanistic data) but lacking RCT validation for non-ejaculatory orgasm outcomes (Level IV).

Edging (orgasm control training): Repeated approach-and-withdrawal from ejaculatory inevitability trains interoceptive awareness and extends pre-climax arousal duration. Evidence supports efficacy for PE management (Level III). Evidence for voluntary nonejaculatory orgasm is weak and confined to self-report series (Level IV/V).

Technique	Physiological Target	Evidence for IELT Extension	Evidence for Non-Ejaculatory Orgasm
Start-stop method	Sensory interruption; arousal reset strong	Level II — Moderate-	Level V — Not applicable
Pelvic floor training BS/LA	tone; proprioception	Level II — Moderate-strong	Level IV — Mechanistically confounded
Mindfulness (MBST)	Performance anxiety; SIS reduction	Level II — Moderate	Level IV — Indirect only
Edging	Interoceptive threshold	Level III — Moderate	Level IV/V — Anecdotal training
Diaphragmatic attenuation	Vagal tone; sympathetic	Level III — Mechanistic	Level IV — No RCT data breathwork
Reverse Kegel	Pelvic floor down-	Level IV — Limited	Level V — Anecdotal training

Perineal (Taoist)	pressure	Mechanical occlusion	ureteral	Level I — Retrograde ejaculation only	Contraindicated epididymitis, pelvic muscle spasm	—	Risk	of
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Table 5. Behavioral techniques: physiological mechanisms and evidence grades for ejaculatory control versus non-ejaculatory orgasm.

Clinical Implications: Behavioral sex therapy has robust evidence for PE management. Evidence for volitional non-ejaculatory orgasm is uniformly weak across all techniques. Perineal pressure techniques carry documented risks and should be explicitly counseled against.

6. Traditional Sexual Practices: Historical Context and Biomedical Evaluation

6.1 Taoist Sexual Cultivation (Yangsheng)

Taoist sexual philosophy, codified in texts including the *Huangdi Neijing* and *Ishimpo* (>2,000 years ago), conceptualized semen as the material manifestation of *Jing*—a finite vital essence whose depletion through ejaculation accelerated aging and disease. Practices aimed to transform *Jing* into *Qi* (vital energy) through breath regulation, meditation, and coital non-ejaculation (*Huan Jing Bu Nao*—"returning the essence to nourish the brain").

Modern biochemistry provides no analog to *Jing* in physiological terms. Semen is a metabolically inexpensive secretion; its production consumes less energy per ejaculate than a brief bout of physical exercise. The *Huan Jing Bu Nao* concept of semen traveling to the brain through energy channels is anatomically unfounded—there exists no physiological pathway connecting the ejaculatory tract to the central nervous system for retrograde fluid or energy transport.

What the Taoist framework did capture, with remarkable intuitive accuracy, is the neurobiological dissociability of orgasm and ejaculation—a distinction not formally characterized in Western medicine until the late 20th century. The error lay in the mechanistic explanation, not the observed phenomenon.

6.2 The "Injaculation" Misconception

Modern popular adaptations (notably Mantak Chia's *The Multi-Orgasmic Man*, 1996) describe "injaculation"—manual perineal compression at the moment of ejaculatory expulsion to redirect semen internally. The anatomical reality is unambiguous: external perineal compression during expulsion forces semen retrograde through the bladder neck via increased urethral backpressure. This constitutes **iatrogenic retrograde ejaculation**, not energy redirection. Repeated forceful retrograde ejaculation against a partially closed sphincter risks ascending infection (epididymitis, prostatitis), high-pressure urethral injury, and pelvic floor muscle spasm.

6.3 Tantric Traditions and Neo-Tantra

Classical Tantra (5th–10th century CE) was a spiritual ritual system in which sexual practices (*Maithuna*) served as vehicles for transcendence rather than physiological optimization. Contemporary "Neo-Tantra" in Western sexology retains practices promoting prolonged non-ejaculatory intercourse, synchronized breathing, and somatic awareness. While Neo-Tantric practice demonstrably reduces performance anxiety and enhances emotional intimacy—outcomes with legitimate psychological evidence—its specific claims of physiological benefit from semen retention lack clinical validation.

6.4 Traditional Claims versus Biomedical Evidence

Traditional Claim	Proposed Mechanism	Scientific Evaluation	Evidence Level
Orgasm can be separated from ejaculation	Pleasure dissociable from fluid emission	Validated — Level I clinical proof	Level I
Ejaculation depletes vital energy	Loss of irreplaceable Jing essence	Unsupported — Semen is metabolically trivial	Level II
Semen retention elevates testosterone indefinitely	HPG axis upregulation	Refuted — Transient Day 7 spike only	Level II
Breathwork aids ejaculatory control	Autonomic modulation	Supported (mechanistic) — Limited RCT data	Level III–IV
Perineal pressure redirects semen to brain	Energy channel redirection	Refuted — Iatrogenic retrograde ejaculation	Level I

Table 6. Traditional claims versus modern biomedical evidence with graded assessment.

7. Semen Retention: Biological Plausibility and Evidence Appraisal

7.1 Spermatogenic Homeostasis and Reabsorption

A common patient misconception is that retained semen accumulates indefinitely, building physiological pressure or altering systemic biochemistry. In reality, the male reproductive tract maintains dynamic homeostasis: continuous spermatogenesis (~74 days) is balanced by epididymal reabsorption of senescent spermatozoa. Aged sperm undergo apoptosis; macrophage phagocytosis recycles constituent amino acids, lipids, and trace minerals into the vascular compartment. Glandular secretions (prostatic, seminal vesicle) undergo mucosal reabsorption. Excess accumulation beyond absorptive capacity may be relieved by nocturnal emission—a physiological, REM-sleep-associated reflex discharge requiring no pathological interpretation.

The clinical implication: semen retention produces no measurable metabolic accumulation, no circulatory change in seminal constituents, and no systemic biochemical alteration distinguishable from the normal fluctuations of the hypothalamic-pituitary-gonadal (HPG) axis.

7.2 Testosterone Dynamics During Abstinence

The endocrine literature on sexual abstinence and testosterone is more nuanced than either retention advocates or critics typically acknowledge.

The most-cited study (Jiang et al., 2003; n=28) demonstrated a transient serum testosterone peak at Day 7 of abstinence reaching approximately 145.7% of baseline ($p < 0.01$), followed by return to baseline by Day 14. This finding—frequently misrepresented online as proof that "NoFap elevates testosterone"—has been replicated in smaller studies but carries important caveats:

1. The sample was small (n=28), with no placebo control for behavioral cofactors.
2. The peak is transient (returns to baseline by Day 14); no study demonstrates sustained supraphysiological testosterone with prolonged abstinence.
3. Endocrine homeostasis via the HPG axis (hypothalamic GnRH pulsatility → pituitary LH → Leydig cell steroidogenesis) is tightly regulated by negative feedback. No mechanism exists by which retention of seminal fluid—produced by the prostate and seminal vesicles, not the testes—could sustainably alter HPG axis setpoints.
4. Studies by Exton et al. (2001) found no significant testosterone differences between abstinent and sexually active conditions in controlled laboratory settings, and conflicting results persist in the literature.

The clinical verdict: a transient Day 7 testosterone fluctuation is real but physiologically modest, mechanistically unexplained, and cannot be extrapolated to support claims of sustained anabolic benefit from long-term semen retention.

7.3 Critical Appraisal of Specific Claims

Physical energy: Post-ejaculatory fatigue is real, mediated by prolactin, oxytocin, and parasympathetic rebound. However, this refractory state lasts hours, not days. The sustained energy improvements reported by retention practitioners are methodologically confounded: NoFap communities typically adopt simultaneous lifestyle changes (improved sleep hygiene, reduced pornography use, increased exercise) that independently account for reported vitality improvements. No controlled study has isolated retained semen volume from lifestyle co-factors as an independent energy variable.

Cognitive performance and "transmutation": The concept of sexual energy "transmutation" into cognitive drive has no mechanistic basis in neuroscience. However, a plausible indirect pathway exists: compulsive pornography use in susceptible individuals may disrupt mesolimbic dopaminergic reward sensitivity via mechanisms analogous to behavioral addiction; discontinuation may partially restore motivational drive. This is a pornography-use effect, not a semen-volume effect.

Athletic performance: Sports medicine literature provides the most methodologically rigorous data. Multiple controlled studies examining pre-competition abstinence versus ejaculation show no significant differences in grip strength, VO2 max, anaerobic power output, or serum anabolic hormones (Boone & Gilmore, 1995; Sztajzel et al., 2000; Level II/III). Current ISSM and sports medicine consensus does not support mandatory precompetition abstinence.

Prostate cancer risk: The relationship between ejaculation frequency and prostate cancer risk is one of the best-evidenced findings in this field, and it directly contradicts retention advocacy. Rider et al. (2016; n=31,925; 18-year follow-up) demonstrated a statistically significant 20% reduction in prostate cancer incidence in men ejaculating ≥ 21 times/month versus those ejaculating 4–7 times/month (HR 0.81; 95% CI 0.72–0.92). The proposed mechanism involves reduced prostatic stagnation and carcinogen clearance with frequent ejaculation. This epidemiological finding (Level II) represents the strongest evidence against prolonged retention for health optimization.

Semen quality: The relationship between abstinence duration and semen parameters follows an inverted-U curve well-characterized in reproductive medicine. Two to five days of abstinence optimizes total sperm count and motility (WHO guidelines recommend 2–5 days for semen analysis). Beyond 7 days, sperm DNA fragmentation increases and motility declines—findings of direct relevance to men using retention practices while attempting fertility (Level I/II; Mayorga-Torres et al., 2015).

Claimed Benefit		Empirical Evidence	Evidence Grade	Clinical Verdict
Elevated testosterone		Transient Day 7 spike only; returns to baseline	Weak — Level IV	Refuted for sustained benefit
Enhanced energy	physical	Confounded by lifestyle co-factors; refractory state resolves in hours	Weak — Level IV	Unsupported as independent effect
Cognitive focus/transmutation		No direct evidence; indirect via reduced pornography use	Limited — Level IV	Confounded; not attributable to semen volume
Athletic performance		No difference in controlled sports medicine trials	Weak — Level II/III	Refuted
Prostate protection	cancer	Epidemiological data favor higher ejaculation frequency	Contradicted — Level II	Refuted; higher frequency is protective
Semen improvement	quality	2–5 day abstinence optimal; >7 days increases DNA fragmentation	Strong — Level I/II	Nuanced: optimal window exists

Table 7. Semen retention health claims: evidence assessment and clinical verdicts.

8. Psychological Mechanisms of Ejaculatory Control

8.1 The Dual Control Model

The Dual Control Model (Bancroft & Janssen, 2000) conceptualizes male sexual response as a dynamic balance between the Sexual Excitation System (SES)—driven by dopaminergic reward, erotic stimuli, and emotional intimacy—and the Sexual Inhibition System (SIS)—driven by serotonergic tone, performance anxiety, threat appraisal, and cultural guilt. This model has significant empirical support from psychophysiological studies (Level II/III).

Ejaculatory timing disorders represent failures of this balance: premature ejaculation reflects SES dominance or SIS deficiency; delayed ejaculation or anejaculation reflects SIS excess. Behavioral techniques for ejaculatory control operate by recalibrating this balance—start-stop and edging by extending SES exposure without reaching SEG threshold; mindfulness by reducing SIS activation; breathwork by attenuating the sympathetic surge that bridges SIS release and sympathetic emission.

8.2 Interoception, Spectatoring, and Attentional Regulation

Spectatoring—the self-conscious, evaluative monitoring of sexual performance during intercourse (Masters & Johnson, 1970)—activates prefrontal threat appraisal circuits, generating cortisol and sympathetic arousal that paradoxically accelerate emission and impair erection. This mechanism explains a clinical paradox: men who desperately try to delay ejaculation often ejaculate faster.

Interoceptive training addresses this by redirecting attentional focus from performance evaluation to somatic body signals: diaphragmatic expansion, peripheral warmth, subtle pelvic muscle tone. Training interoceptive sensitivity allows earlier recognition of preinevitability arousal signals, providing a wider window for voluntary modulation. This principle—not a specific technique, but an attentional strategy—underlies the partial efficacy of edging, mindfulness, and breathwork.

The psychological literature has not yet clearly established whether voluntary nonejaculatory orgasm in healthy males requires merely interoceptive skill, or whether it requires a fundamental neurobiological predisposition (e.g., baseline SEG threshold, individual variation in descending inhibitory tone). This question remains unanswered.

9. Methodological Appraisal and Research Gaps

9.1 Limitations of Existing Literature

The evidence base for non-ejaculatory orgasm and semen retention is characterized by systematic methodological weaknesses that preclude strong conclusions:

Evidence hierarchy imbalance: The majority of published literature on voluntary nonejaculatory orgasm consists of Level V–VII evidence—case reports, narrative reviews, online forum surveys, traditional texts, and self-help books. High-grade RCTs are absent. Even for established behavioral techniques (start-stop, PFMT), most trials specifically address premature ejaculation rather than the distinct question of non-ejaculatory orgasm.

Definitional heterogeneity: Across published studies, terminology for "dry orgasm," "anejaculation," "retrograde ejaculation," "multiple orgasm," and "voluntary non-ejaculatory orgasm" is used inconsistently and often interchangeably. Without standardized definitions, cross-study synthesis is methodologically compromised.

Selection and publication bias: Self-selected participants from retention communities represent cohorts with strong confirmation bias and expectancy effects. Positive experiences are disproportionately reported; negative or null findings are underreported. Online forum data cannot be treated as observational evidence.

Absence of objective verification: No published study has verified voluntary nonejaculatory orgasm using concurrent pelvic EMG (to confirm orgasmic contractions), postorgasm bladder urinalysis (to rule out retrograde ejaculation), and validated subjective orgasm intensity scales in healthy male volunteers. Without such multi-modal verification, self-reported non-ejaculatory orgasm cannot be distinguished from partial arousal, anorgasmic ejaculation, or retrograde ejaculation.

Confounding in semen retention studies: Studies assessing semen retention benefits virtually always involve simultaneous behavioral changes (pornography cessation, improved sleep, exercise adoption) that independently produce the reported outcomes. No study has isolated fluid retention from cofactors using appropriate controls.

9.2 Priority Research Agenda

Research Priority	Methodology	Primary Outcome
Neuroimaging of voluntary nonejaculatory orgasm	fMRI + concurrent pelvic EMG in trained cohorts	Cortical activation patterns vs. standard orgasm
Objective verification protocol	Post-orgasm urinalysis + EMG + IIEF scales	Confirm true non-ejaculatory orgasm vs. retrograde
Multi-arm behavioral RCT	Start-stop vs. PFMT vs. MBST vs. sham; n≥100/arm	IELT, non-ejaculatory orgasm frequency
Endocrine longitudinal profiling	Serial testosterone, LH, prolactin, cortisol; 30-day controlled abstinence	HPG axis dynamics; confounders controlled
Prostate health and ejaculation frequency	Prospective cohort; PSA, biopsy endpoints	Optimal ejaculation frequency for cancer prevention

Table 8. Priority research agenda for voluntary non-ejaculatory orgasm and semen retention.

10. Discussion and Clinical Implications

10.1 Integrative Synthesis

This review establishes several conclusions with differing levels of certainty:

Established with strong evidence (Level I/II): Orgasm and ejaculation are neurobiologically distinct. The SEG (L3–L5) orchestrates ejaculation autonomously once its threshold is reached; the cortical orgasm network operates in parallel. Clinical models—prostatectomy, retrograde ejaculation, SCI—provide definitive proof that pleasurable climax occurs in the complete absence of semen expulsion. These facts are not contested in the scientific literature.

Established with moderate evidence (Level II/III): Behavioral techniques (start-stop, PFMT, MBST) produce clinically meaningful improvements in ejaculatory latency in men with PE. The serotonergic mechanism of SSRIs for PE delay is well-characterized. Ejaculation frequency ≥ 21 /month is associated with reduced prostate cancer risk. Abstinence >7 days impairs semen quality.

Unproven (Level IV/V): Volitional non-ejaculatory orgasm through behavioral training in healthy males. No RCT, no prospective cohort study, and no laboratory-verified case series has established that healthy men can reliably acquire this capacity through any behavioral technique.

Refuted (Level I/II): Sustained testosterone elevation from semen retention. Perineal pressure as a mechanism for semen "retention" (it is retrograde ejaculation). Athletic performance enhancement from pre-competition abstinence.

10.2 Clinical Practice Recommendations

Urologists, sex therapists, endocrinologists, and primary care clinicians will increasingly encounter patients presenting with questions about non-ejaculatory orgasm, semen retention, and their health consequences. Evidence-based counseling should address:

1. **Physiological reassurance:** Orgasm and ejaculation are neurobiologically independent. Dry orgasm following prostatectomy or with alpha-blocker therapy is physiologically normal and preserves subjective pleasure. Patients should be counseled without stigma.
1. **Harm reduction:** Manual perineal compression during ejaculation (the "injaculation" technique) causes iatrogenic retrograde ejaculation and risks epididymitis, prostatitis, and pelvic floor spasm. This should be explicitly advised against.

1. **Semen retention counseling:** The transient Day 7 testosterone spike does not translate to sustained anabolic benefit. Men seeking retention for fertility should be informed of the optimal 2–5 day abstinence window; >7 days is counterproductive for sperm DNA integrity.
1. **Behavioral therapy referral:** For men with PE or seeking improved ejaculatory control, referral to structured PFMT, start-stop training, or MBST with a qualified sex therapist is appropriate, with realistic expectations communicated regarding timelines and outcomes.
1. **Autonomy with clarity:** Patients practicing semen retention for personal, spiritual, or cultural reasons should have their autonomy respected while receiving accurate evidence-based information about testosterone, athletic performance, and prostate health.

10.3 Conclusions

Male orgasm and ejaculation are dissociable physiological phenomena, governed by distinct but interacting neural circuits. This is no longer a theoretical proposition—it is clinically established through multiple Level I/II models in urology and neurology. The central unresolved question is whether this dissociation is reliably achievable in healthy males through volitional behavioral training, and current evidence does not support an affirmative answer at a level of rigor required for clinical recommendation.

The discrepancy between the biological plausibility of voluntary non-ejaculatory orgasm and its empirical validation gap is this field's defining research challenge. Rigorous, multimodal, laboratory-verified RCTs—combining neuroimaging, pelvic EMG, post-orgasm urinalysis, and validated subjective measures—are required before clinicians can offer evidence-based guidance on behavioral acquisition of non-ejaculatory orgasm in healthy populations. Until such evidence exists, clinicians should distinguish what is biologically demonstrable from what remains traditionally asserted, culturally believed, or anecdotally reported.

Conflicts of Interest

The authors declare no conflicts of interest.

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